

HARSH SAHU

Python Backend Developer | AI Engineer

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PROFESSIONAL SUMMARY

Python Backend Engineer with 1.5+ years of experience building scalable, real-time AI systems — including production voice AI pipelines (Pipecat, multi-provider STT/TTS/LLM orchestration), RAG pipelines, LangGraph-based agentic workflows, and async REST APIs using FastAPI, Django/DRF, SQLAlchemy, and LangChain. Skilled in PostgreSQL, Redis, Celery, and OpenAI/LLM provider integrations, with a track record of reducing latency, fixing production-grade pipeline bugs, and optimizing async architectures at scale. Currently targeting AI Engineer / AI-GenAI Backend Engineer roles.

TECHNICAL SKILLS

- **Languages:** Python 3, SQL
- **Frameworks:** FastAPI, Django, Django REST Framework (DRF), LangChain, LangGraph, Pipecat
- **Databases & ORM:** PostgreSQL, Redis, Qdrant (Vector DB), SQLAlchemy (async/asyncpg), Alembic
- **AI / ML:** OpenAI API, multi-provider LLM/STT/TTS integration (DeepSeek, Perplexity, Grok), RAG Pipelines, Embeddings, Pydantic
- **Real-Time Systems:** WebSockets, Django Channels, Async Programming, Celery, real-time voice/telephony pipelines
- **Testing & Tools:** Pytest, Postman, Git, GitHub, Docker, VS Code, PyCharm
- **Concepts:** REST API Design, OOP, DBMS, System Design, Linux, JWT Auth

WORK EXPERIENCE

Software Development Engineer I (SDE-1) — ScaleReal Technologies Pvt. Ltd.

Aug 2025 – Present

- Promoted from intern to SDE-1 within 6 months based on performance and delivery.
- Built and maintain a production voice AI platform using Pipecat, FastAPI, and WebSockets, integrating multiple LLM, STT, and TTS providers for real-time conversational pipelines with telephony support.
- Diagnosed and resolved critical pipeline bugs across the provider stack — including silent STT failures from VAD misconfiguration, TTS audio drop-outs from WebSocket race conditions, and LLM error-propagation gaps.
- Built 15+ production REST APIs serving low-latency responses (<300ms) using Django REST Framework.
- Designed a real-time trading system using Django Channels and WebSockets for concurrent market updates with minimal lag.
- Built a RAG pipeline with LangChain and Qdrant vector DB for document ingestion and context-aware responses, improving answer accuracy by ~35%.
- Reduced PostgreSQL query response time by 40% through index optimization, async query restructuring, and N+1 elimination using SQLAlchemy.

Python Backend Intern — ScaleReal Technologies Pvt. Ltd.

Jan 2025 – Jul 2025

- Developed REST APIs using DRF for production use cases with full test coverage via Pytest.
- Built a real-time notification system using Redis pub/sub and async processing, reducing delivery latency across 500+ concurrent users.
- Wrote unit and integration tests using Pytest, achieving 80%+ code coverage on assigned modules.

PROJECTS

RAG SaaS Chatbot Platform — Python, Django, DRF, LangChain, Qdrant, OpenAI, Redis [\[Link\]](#)

- Designed a multi-tenant SaaS platform with role-based access control (RBAC) and per-tenant knowledge base management.
- Built a retrieval pipeline using LangChain + Qdrant for semantic, context-aware chatbot responses.
- Implemented secure API key authentication for third-party integrations; published Swagger and Postman docs.
- Enabled token-level streaming responses, reducing perceived latency by ~50%.

RAG-Based Portfolio Assistant — DRF, Qdrant, OpenAI Embeddings, LangGraph, Render, Supabase [\[Link\]](#)

- Built a portfolio chatbot using OpenAI embeddings + Qdrant for semantic search over personal project data.
- Built a document ingestion and retrieval pipeline with session-based memory using LangGraph, Redis, and Qdrant.
- Deployed a scalable backend on Render; integrated Supabase for persistent storage.

E-Commerce Platform — Django, Django Channels, Redis, Bootstrap, Google OAuth [\[Link\]](#)

- Implemented Google OAuth and JWT-based secure authentication with password recovery.
- Built live customer support chat using Django Channels (WebSockets) and Redis as the channel layer.
- Designed an order tracking system with product search and filtering by category, name, and description.

EDUCATION

B.Tech, Computer Science & Engineering

2022 – 2025

Dr. A.P.J. Abdul Kalam Technical University — CGPA: 8.0 / 10

CERTIFICATIONS & ACHIEVEMENTS

- **Certifications:** [Introduction to MCP, SQL Bootcamp, Python & Django Framework, Hugging Face Agents](#)
- Solved 150+ DSA problems on LeetCode; earned a 50-day coding streak badge.
- Built a LinkedIn audience of 9,000+ followers by consistently sharing Python and backend engineering content.